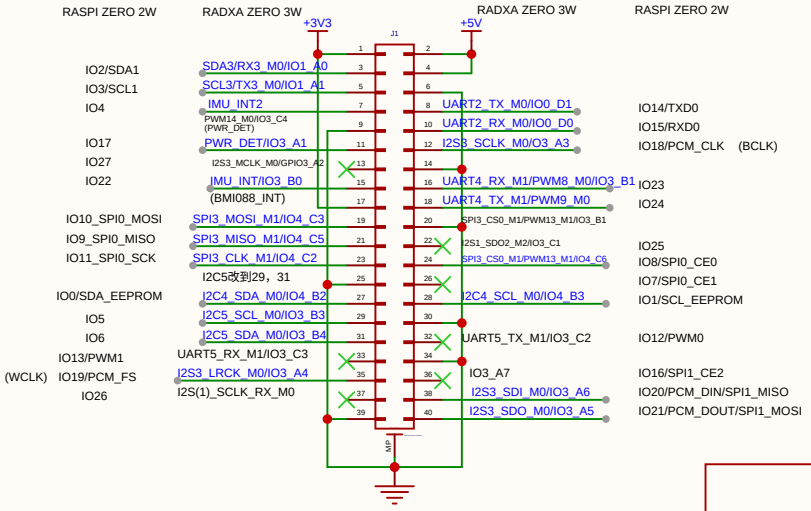
\"RPI_Robot_HAT\" is a raspberry PI HAT designed for small/medium size robots. It has an IMU, Dynamixel connectors and audio. Its dimensions fits a raspberry PI zero

| Release | Date | Designer | Check | Comments |
|---|---|---|---|---|
| A1 | 06/03/2026 | EB | SN | Initial drawing |
| B1 | 06/05/2026 | EB | SN | [fixed] Acc. is at 0x19 I2C address<br>[fixed] Soft reset of the audio DAC<br>[fixed] Ideal diode<br>[fixed] Know position of IMU<br>[update] Dynamixel_dir is based on Tx<br>[added] IMU can be disconnected<br>[added] IMU_IT is cabled<br>[added] Access to I2C0<br>[added] Switch off detection |
| C1 | 08/07/2026 | EB | SN | [fix] 3V3 on IO10/11<br>[fix] MOS direction of Ideal diode<br>[update] I2C3 is by default instead of reserved I2C |

Cabled buses on Raspberry PIs
- PI zero 2W:
  - I2C0 (  pin 27/28 ID_SD/C): reserved, but can be on J6 with 0R
  - I2C1 (IO  2/ 3 - pin  3/ 5 SDA/SCL): IMU + Audio + Qwiic J5
- PI 4:
  - same I2C0/1
  - I2C3 (IO  4/ 5 - pin  7/29 SDA/SCL): Qwiic J6
  - I2C4 (IO  8/ 9 - pin 24/21 SDA/SCL): Qwiic J7
  - I2C5 (IO 10/11 - pin 19/23 SDA/SCL): Qwiic J8
- PI 5:
  - same I2C0/1
  - I2C2 (IO  2/ 3 - pin  3/ 5 SDA/SCL): Qwiic J5

Cabled GPIOs on the 40-way Header:
- GPIO00/01: HAT identification over I2C (reserved)
- GPIO02: IMU + Audio + Stemma/QT - SDA
- GPIO03: IMU + Audio + Stemma/QT - SCL
- GPIO04: Stemma/QT - SDA
- GPIO05: Stemma/QT - SCL
- GPIO06: Switch off detection
- GPIO07
- GPIO08: Stemma/QT - SDA
- GPIO09: Stemma/QT - SCL
- GPIO10: Stemma/QT - SDA
- GPIO11: Stemma/QT - SCL
- GPIO12
- GPIO13:
- GPIO14: Dynamixel - UART_Tx
- GPIO15: Dynamixel - UART_Rx
- GPIO16
- GPIO17
- GPIO18: Audio, PCM Clk
- GPIO19: Audio, PCM FS
- GPIO20: Audio, Din
- GPIO21: Audio, Dout
- GPIO22: IMU_IT
- GPIO23
- GPIO24
- GPIO25
- GPIO26
- GPIO27

# 增加风扇接口



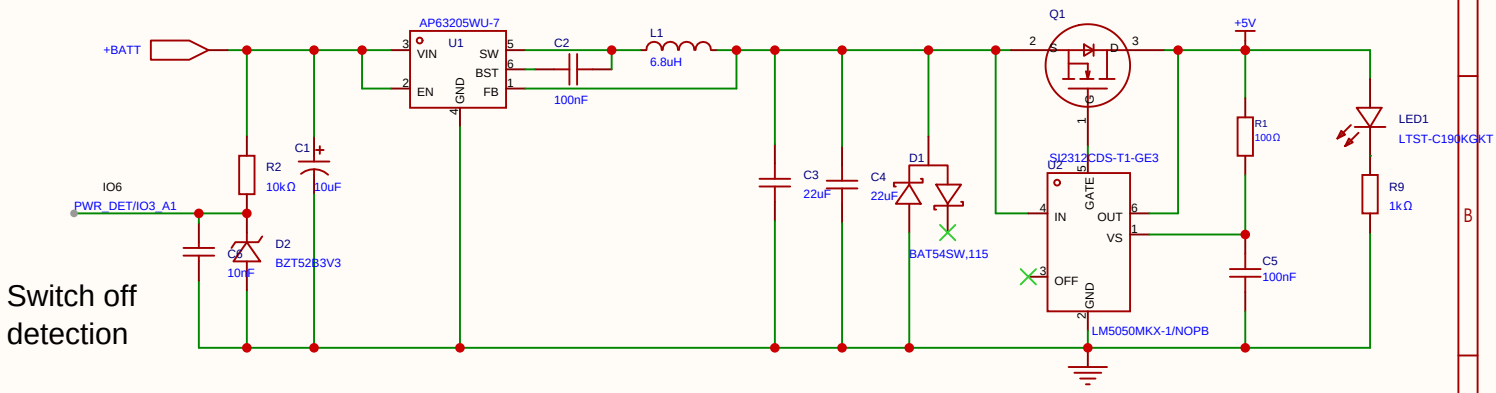

Fiducials



Sheet : /
File : elec_RPI_Robot_HAT

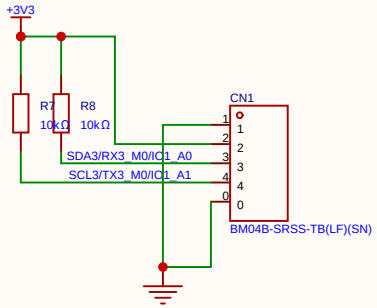
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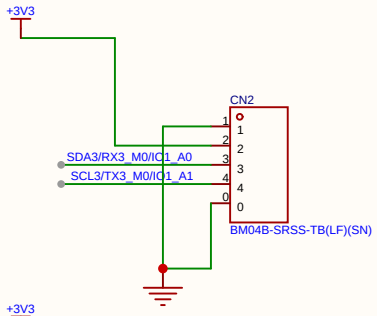
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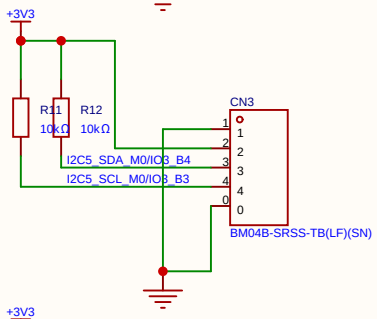


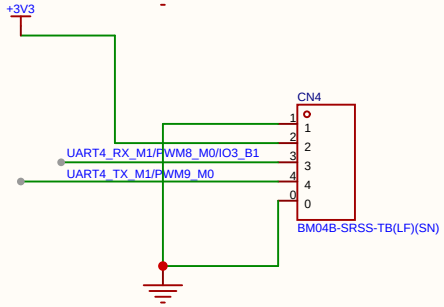

| | | | |
|---|---|---|---|
| 原理图 | Microduck-Hat-for-Radxa-Zero_2 | 创建日期 | 2026-10-07 |
| 板子 | Microduck-Hat-for-Radxa-Zero_fetech-v1.3_1 | 更新日期 | 2026-10-07 |
| 绘制 | | 图页 | Power |
| 审阅 | | Microduck-Hat | |
| | | 版本 | 尺寸 | 页 2 共 5 |
| 嘉立创EDA | | V1.0 | A4 | 嘉立创EDA |

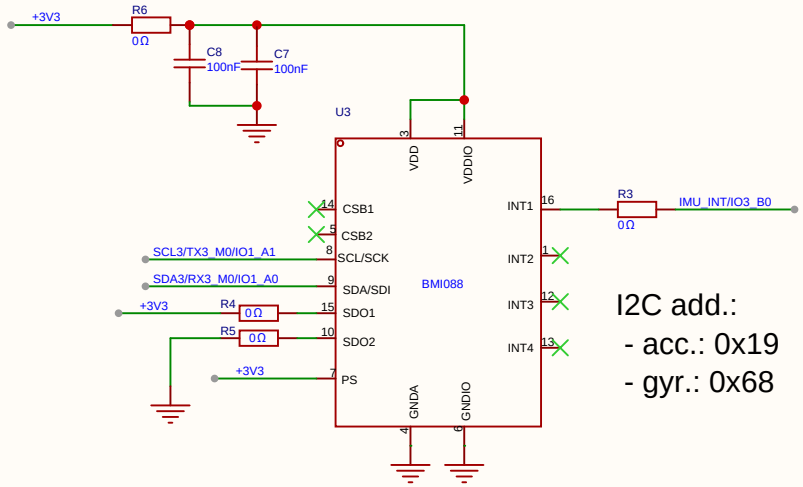
# I2C SENSOR BUS



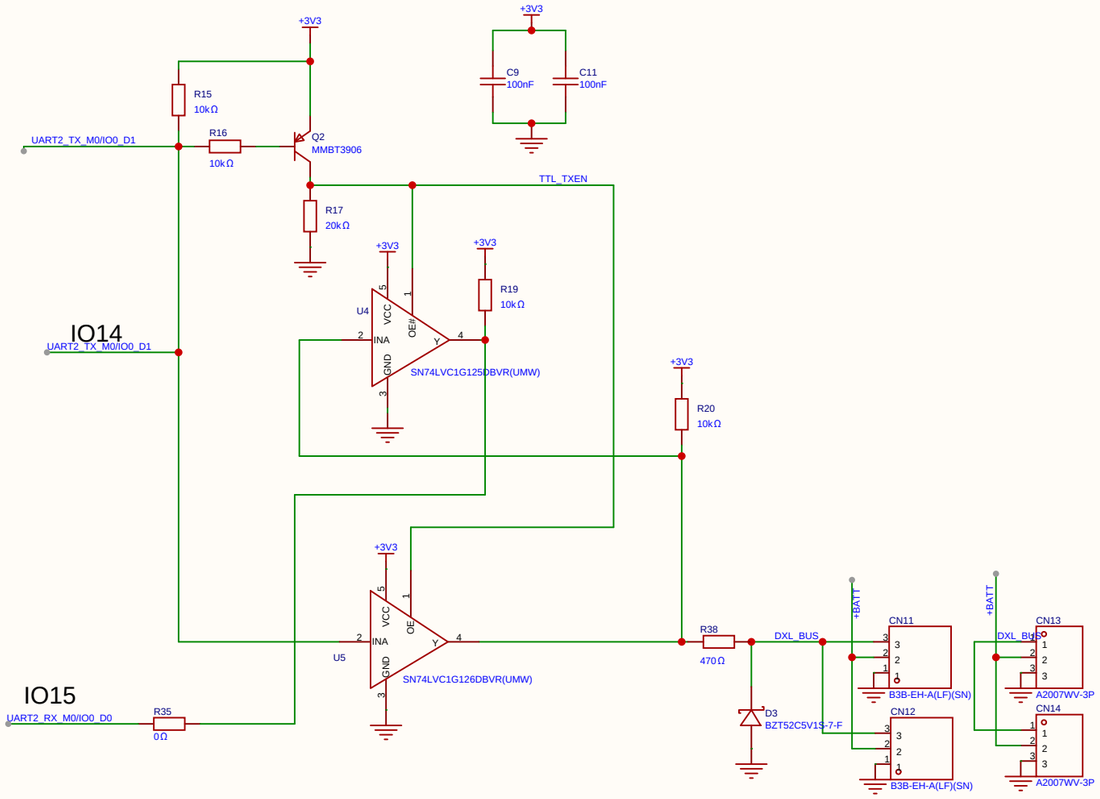



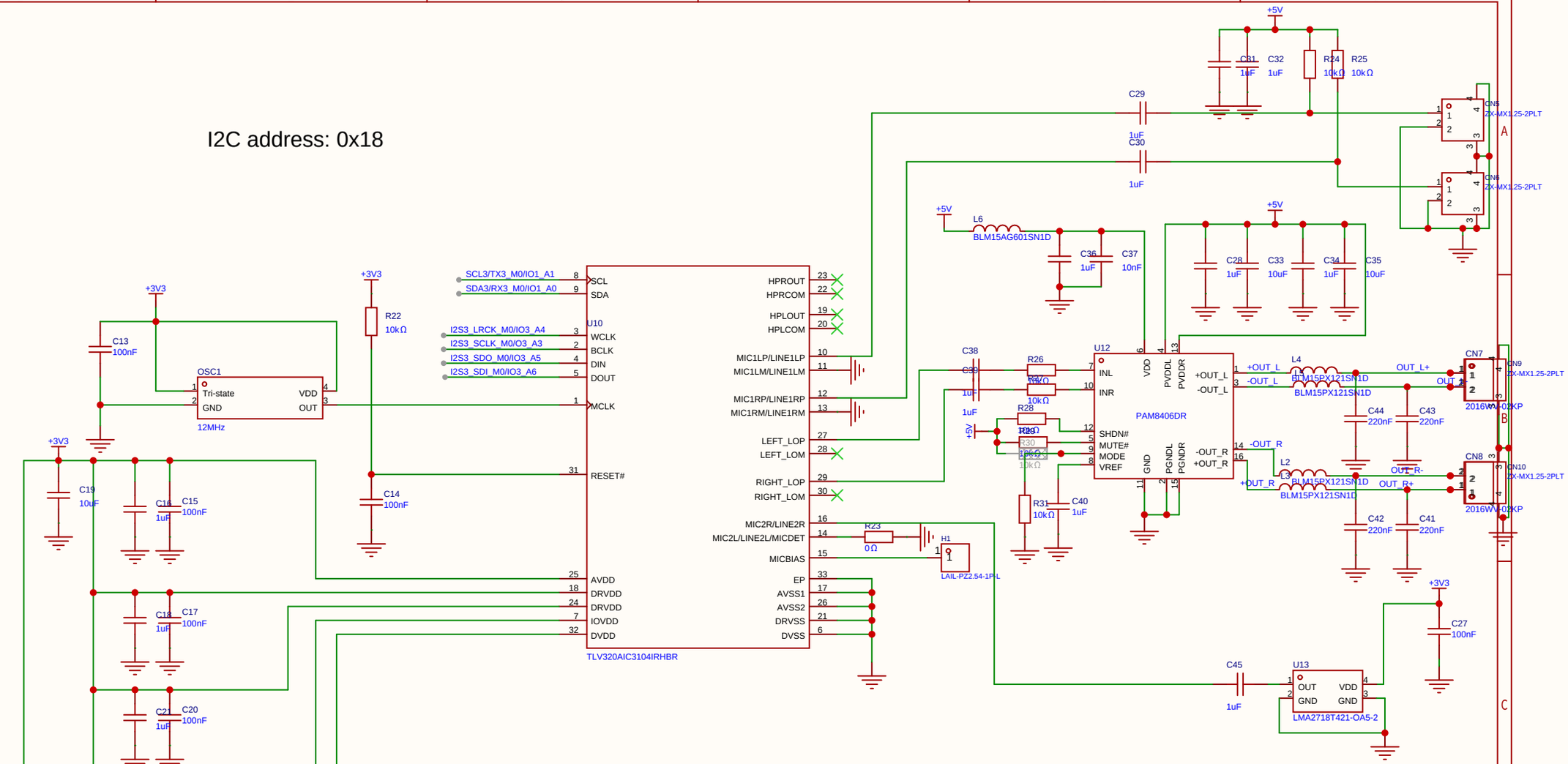

I2C add.:
- acc.: 0x19
- gyr.: 0x68

| | | | |
|---|---|---|---|
| 原理图 | Microduck-Hat-for-Radxa-Zero_2 | 创建日期 | 2026-10-06 |
| | | 更新日期 | 2026-10-07 |
| 板子 | Microduck-Hat-for-Radxa-Zero_fetech-v1.3_1 | 图页 | Sensor |
| 绘制 | | | |
| 审阅 | | Microduck-Hat | |
| | | 版本 | 尺寸 |
| 嘉立创EDA | | V1.0 | A4 |
| | | 页 3 共 5 | |
| | | 嘉立创EDA | |

# SERVOS




| | | | |
|---|---|---|---|
| 原理图 | Microduck-Hat-for-Radxa-Zero_2 | 创建日期 | 2026-10-06 |
| | | 更新日期 | 2026-10-07 |
| 板子 | Microduck-Hat-for-Radxa-Zero_fetech-v1.3_1 | 图页 | Servo |
| 绘制 | | | |
| 审阅 | | Microduck-Hat | |
| | | 版本 | 尺寸 |
| | | V1.0 | A4 |
| 嘉立创EDA | | 页 4 共 5 | |
| | | 嘉立创EDA | |




| | | | |
|---|---|---|---|
| 原理图 | Microduck-Hat-for-Radxa-Zero_2 | 创建日期 | 2026-10-06 |
| 板子 | Microduck-Hat-for-Radxa-Zero_fetech-v1.3_1 | 更新日期 | 2026-10-20 |
| 绘制 | | 图页 | Audio |
| 审阅 | | Microduck-Hat | |
| | | 版本 | 尺寸 |
| | | V1.0 | A4 |
| | | 页 5 共 5 | |
| 嘉立创EDA | | 嘉立创EDA | |